Fiscal Year:	FY 2026	**Task Last Updated:**	FY 08/02/2025
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Project Title:	Teleportation of Atomic States Between Earth and Space		
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Human Research Program Elements:	None		
Human Research Program Risks:	None		
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Comments:			
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Task Description:	This task aims to investigate the feasibility of future atomic-state teleportation between Earth and space. The key focus lies on remotely entangling atom-like qubits over turbulent free-space optical channels. A successful demonstration of this experiment will mature the technology for space quantum networks and inform future research directions for remotely entangling atom-like quantum states. While quantum-atomic systems hold great promises as future sensors and network nodes, many underlying technologies remain at low maturity for deployment in space or across large-scale networks. In this task, we will remotely entangle atomic-like states in semiconductor hosts via free-space optical channels representative of future Earth-to-space links. This effort will advance technology maturity and improve key performance metrics of entanglement systems, benefiting all other efforts to develop space-based quantum-atomic capabilities. Atomic teleportation is the technique by which an atom's quantum state is transferred to another atom. This process underpins proposed networked quantum systems such as quantum clock networks, quantum telescopes and distributed quantum information systems. It also provides a gateway to testing foundational physics involving the coupling of gravitation and quantum measurement. Realizing atomic teleportation requires a synthesis of atomic physics and quantum-optical entanglement - two core areas of interest for the NASA Science Mission Directorate's Biological and Physical Sciences (SMD-BPS) Fundamental Physics program. By the conclusion of the task, the NASA Science Mission Directorate will have advanced atomic teleportation technology and identified the next research directions needed to demonstrate remote entanglement with space-based nodes.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Successful implementation of the envisioned experiment provides several tangible benefits to the field of quantum communications: - Quantum communications across free-space optical channels: experimental implementation of atomic teleportation requires advanced high-gain free-space-optical link systems, including rejection of atmospheric turbulence, high-precision synchronization, and phase stabilization. Such optical links have immediate applications for other quantum communication systems, such as photonic quantum teleportation, or quantum telescopes. To support future quantum science and engineering goals, the free-space optical links that are developed as part of this task may serve as a testbed for quantum-communication experiments across turbulent channels. - Improved coupling from atomic qubits: enhancing the coupling efficiency of qubits in solid-state hosts to free space has broad applications beyond the immediate task, for example for networked quantum computers or sensors. Any steps towards this goal may be transferable to other atom-like systems. - Tests of quantum field theory in curved spacetime: recent studies have proposed using remotely entangled atom-like qubits located in different gravitational potentials for novel experiments at the interface of quantum mechanics and general relativity. The experiment developed in this task will be suitable for terrestrial tests in the near future.
	Remotely entangling atom-like qubits, over free-space optical links, requires several components operating near the state-of-the-art: • A qubit platform with individually addressable spin states. We use ytterbium ions in yttrium orthovanadate (YVO4) hosted in nanophotonic cavities as a starting point. From a small ensemble of ions, individual ground-state spin states can be controlled and read out due to their spectral distinguishability. The ions have narrow linewidths on the order of MHz and emit in the near-infrared, around 984 nm. Within this task, we investigate multiplexing of several ions to boost entanglement rates, as well as replacing the host material to improve coupling into free space and the emitter's spectral properties. • An entanglement protocol. To remotely entangle two or more atom-like qubits, a joint measurement of their emitted photons is performed at a central measurement station. Two entanglement protocols have been previously developed and may be investigated across free-space optical links. The "two-photon" protocol requires two-photon interference of photons emitted from two qubits to facilitate the entanglement. The rate of entanglement events scales with the product of the link efficiencies between the qubits and the measurement, which greatly limits the suitability for long distance links. In contrast, the "single-photon" protocol scales with the efficiency of a single link, at the cost of populating only a small fraction of emission time slots. Moreover, the single-photon protocol requires optical phase stabilization. Within this task, we aim to implement both protocols to trade off their strengths and weaknesses. • Free-space optical (FSO) links. A critical component of the experiment: the FSO links that connect the qubits and the measurement station. We envision 10-km-scale FSO links between Caltech and JPL, with both emitters located at the California Institute of Technology (Caltech) and the measurement performed at NASA Jet Propulsion Laboratory (JPL). The FSO links will need to couple light between two single-mode fibers through a turbulent channel with high efficiency on the order of 10% (-10 dB). For the single-photon protocol, the FSO links need to be phase-stabilized on the order of a fraction of the optical wavelength (~100 nm). • Quantum frequency conversion/QFC (optional). To mitigate the impact of atmospheric turbulence, and enable optimal propagation in optical fiber in the future, the frequency of the emitted photons may be converted from the near infrared into the telecommunication C-band. QFC can be achieved by difference-frequency generation (DFG) from a 2700-nm pump laser. The entanglement protocols work as follows: two remotely located ions are each prepared in a local, balanced or unbalanced (depending on the protocol) superposition state and conditionally pumped to an optical excited state. If a photon is emitted, it is sent to the QFC device to convert its wavelength from 984 nm to 1550 nm. After conversion, it is sent over the FSO link and is detected with low probability at the central station. The detection event is timestamped and communicated without delay to both transmitter sites where the ions are located. After local corrections at each ion, the entangled state is generated and can be confirmed by reading out the ions and correlating the measurement outcomes. As evident from this sequence, several additional supporting components are required, such as real-time communication, and phase-control across several components. We made significant progress across all technical components and performed several proof-of-concept experiments to

Task Progress:	guide the development. Theory and model development: To evaluate experimental feasibility and derive requirements for the components, we developed a parametric model built on existing theoretical frameworks and evaluated the key performance parameters, entanglement rate and quantum state fidelity, in the presence of increased loss, phase instability, and background counts. The parametric model was then used to define a set of starting requirements that were used to guide the development of all subsystems and to define a series of critical precursor measurements. Qubit platform: The task supported research aimed at improving the material properties of the host material, as well as improving the rate of emitted photons. Starting with previously achieved entanglement generation between two qubits within the laboratory, multiplexing was demonstrated with the same platform. Multiplexing two qubits yielded a 1.9x increase in the effective entanglement rate. This is a critical result for future space-based links, which will require multiplexing of several qubits to overcome large losses and long propagation delays associated with ground-to-satellite links. To investigate improved qubit properties, yttrium orthovanadate was replaced by CaWO4 as the host material. First results from spectroscopy and fabrication show improved optical properties, long spin coherence times, as well as high-quality factor nanophotonic cavities. These results indicate the suitability of CaWO4 as a future host material for the entanglement system. FSO system: To enable high-gain optical links over the 10-km channel, we first performed characterization of atmospheric turbulence across the free-space optical channel between Caltech and JPL. This is crucial, as atmospheric turbulence is heavily impacted by the local environment, which is especially true for horizontal atmospheric channels in urban areas. We monitored the angle-of-arrival fluctuations of light emitted from Caltech by two light sources (980 nm, 1064 nm) and detected at JPL over several months. Assuming that the system only operates at nighttime, we find that approximately 50% of nights showed manageable turbulence for extended periods of time, with a Fried parameter at 1550 nm exceeding 8 cm. Moreover, the measurements will enable us to predict turbulence strength in the future. In parallel, we assembled a prototype optical-link system with identical, 8-inch transmit and receive telescopes. We tested alignment, pointing, and coupling into single-mode fiber across the turbulent channel between JPL and Caltech and achieved close to 1% end-to-end throughput, consistent with our expectations. We further measured negligible optical background power from light pollution at 1550 nm. These results allowed us to design a pair of transmitter and receiver terminals that couple light from Caltech to JPL into a single-mode fiber with near-optimal efficiency. To avoid higher-order adaptive optics, we designed the system to operate in confocal configuration, which is suitable for the mild turbulence measured. The system is currently being assembled and the path for the quantum channel was completed. We find an end-to-end efficiency of approximately -6 dB in the lab, indicating that the target transmittance of -10 dB across the FSO link is realistic. We expect testing of the high-gain links across the Caltech-to-JPL channel before October. QFC system: Converting the qubit photon wavelength to the C-band requires QFC via DFG with a 2700-nm pump. We developed a model to estimate the performance parameters of the conversion process and used it to design a QFC system. The first prototype of the system was assembled in the laboratory using a periodically poled bulk lithium niobate crystal. While the expected conversion efficiency is low, it will be used to confirm the parametric model and test the components. Once frequency conversion is detected and the parametric models are validated, we will replace the bulk crystal with a waveguide that is estimated to achieve conversion efficiency between 40-60%. Beyond the main components, we started implementation of a real-time feedback communication channel that will transmit detection time tags from JPL to the qubits at Caltech for active phase correction. We further demonstrated phase stabilization of a time-bin state at 10-kHz control-loop frequency, which may be suitable to stabilize the optical phase of the link.
Bibliography Type:	Description: (Last Updated: 08/13/2025)
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Articles in Other Journals or Periodicals	Tiranov A, Green E, Hermans S, Liu E, Chiossi F, Serrano D, Loiseau P, Kumar AM, Bertaina S, Faraon A, Goldner P. "Sub-second spin and lifetime-limited optical coherences in 171Yb3+:CaWO4." arXiv preprint server. Posted April 2, 2025. https://doi.org/10.48550/arXiv.2504.01592 , Apr-2025